

OptiCrop: Smart Agricultural Production Optimization Engine

No. of Functional Features Included in the Solution

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Secure login and registration for users.	Authentication	Done	High
2	Crop Recommendation	Predicts the most suitable crop based on soil and environmental parameters	Machine Learning	Done	High
3	Crop Recommendation	Suggests the appropriate fertilizer for the selected crop.	Recommendation Engine	Done	Medium
4	Weather Information	Displays weather details to assist farming decisions.	Weather Module	Done	Medium
5	User Dashboard	Provides a simple interface to access all project features.	Frontend (UI)	Done	Medium
6	Database Management	Stores user information and prediction history.	MySQL Database	Done	Medium
7	Prediction History	Allows users to view previous crop recommendations.	Database Module	In progress	Medium
8	Market Price Prediction	Predicts crop market prices for better planning.	Analytics Module	pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	6

6Additional / Nice-to-Have Features	3
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Login Page, Dashboard
2	Backend / Logic	3	Crop Prediction, Fertilizer Recommendation, Prediction Processing
3	Database / Storage	2	User Data, Prediction History
4	API / Integration	1	Weather API Integration
5	Security / Authentication	1	User Login & Registration